

ICAD RT-4 · Set 1 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

Name

Roll No

Date

Physics

1	☐ a	☐ b	☐ c	☐ d
2	☐ a	☐ b	☐ c	☐ d
3	☐ a	☐ b	☐ c	☐ d
4	☐ a	☐ b	☐ c	☐ d
5	☐ a	☐ b	☐ c	☐ d
6	☐ a	☐ b	☐ c	☐ d
7	☐ a	☐ b	☐ c	☐ d
8	☐ a	☐ b	☐ c	☐ d
9	☐ a	☐ b	☐ c	☐ d
10	☐ a	☐ b	☐ c	☐ d
11	☐ a	☐ b	☐ c	☐ d
12	☐ a	☐ b	☐ c	☐ d
13	☐ a	☐ b	☐ c	☐ d
14	☐ a	☐ b	☐ c	☐ d
15	☐ a	☐ b	☐ c	☐ d
16	☐ a	☐ b	☐ c	☐ d
17	☐ a	☐ b	☐ c	☐ d
18	☐ a	☐ b	☐ c	☐ d
19	☐ a	☐ b	☐ c	☐ d
20				

Numerical — write the value

21	
22	
23	
24	
25	

Chemistry

1	☐ a	☐ b	☐ c	☐ d
2	☐ a	☐ b	☐ c	☐ d
3	☐ a	☐ b	☐ c	☐ d
4	☐ a	☐ b	☐ c	☐ d
5	☐ a	☐ b	☐ c	☐ d
6	☐ a	☐ b	☐ c	☐ d
7	☐ a	☐ b	☐ c	☐ d
8	☐ a	☐ b	☐ c	☐ d
9	☐ a	☐ b	☐ c	☐ d
10	☐ a	☐ b	☐ c	☐ d
11	☐ a	☐ b	☐ c	☐ d
12	☐ a	☐ b	☐ c	☐ d
13	☐ a	☐ b	☐ c	☐ d
14	☐ a	☐ b	☐ c	☐ d
15	☐ a	☐ b	☐ c	☐ d
16	☐ a	☐ b	☐ c	☐ d
17	☐ a	☐ b	☐ c	☐ d
18	☐ a	☐ b	☐ c	☐ d
19	☐ a	☐ b	☐ c	☐ d
20	☐ a	☐ b	☐ c	☐ d

Numerical — write the value

21	
22	
23	
24	
25	

Mathematics

1	☐ a	☐ b	☐ c	☐ d
2	☐ a	☐ b	☐ c	☐ d
3	☐ a	☐ b	☐ c	☐ d
4	☐ a	☐ b	☐ c	☐ d
5	☐ a	☐ b	☐ c	☐ d
6	☐ a	☐ b	☐ c	☐ d
7	☐ a	☐ b	☐ c	☐ d
8	☐ a	☐ b	☐ c	☐ d
9	☐ a	☐ b	☐ c	☐ d
10	☐ a	☐ b	☐ c	☐ d
11	☐ a	☐ b	☐ c	☐ d
12	☐ a	☐ b	☐ c	☐ d
13	☐ a	☐ b	☐ c	☐ d
14	☐ a	☐ b	☐ c	☐ d
15				
16	☐ a	☐ b	☐ c	☐ d
17				
18				
19				
20				

Numerical — write the value

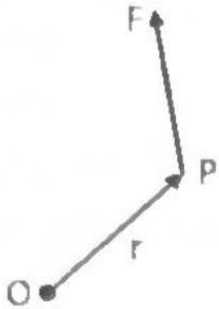
21	
22	
23	
24	
25	

Bubbles — single correct · Box — write the value · each subject out of 100

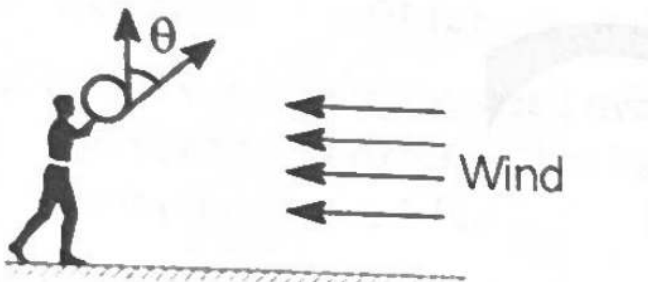
Physics

PART A · Q1–20 · SINGLE CORRECT

1. P2L1V3Q1 A function changes from $y = 2$ to $y = 5$ over $x = 1$ to $x = 3$. Average rate of change? (a) 2 (b) 1.5 (c) 3 (d) 0.5
2. P3L1V3Q5 $0.4\hat{i} + 0.8\hat{j} + c\hat{k}$ represents a unit vector when 'c' is (a) -0.2 (b) $\sqrt{0.2}$ (c) $\sqrt{0.8}$ (d) 0
3. P3L6L7V5Q6 A force of $(4\hat{i} + 3\hat{j} + 2\hat{k})$ N acts at point P(3,4,5) m. Find the torque of the force about point O(1,2,3) m. (a) $(-2\hat{i} + 4\hat{j} - 2\hat{k})$ N-m (b) $(2\hat{i} - 4\hat{j} + 2\hat{k})$ N-m (c) $(-4\hat{i} + 2\hat{j} - 2\hat{k})$ N-m (d) Zero

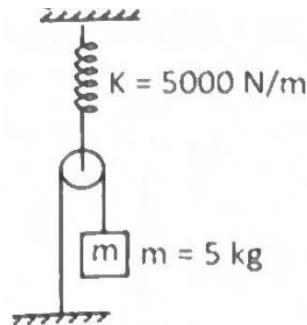


4. P4L1V3Q2 A boat travels 50 km east, then 120 km north, then returns to the starting point by the shortest path; total journey time is 3 h. Find the average velocity over the entire trip. (a) 0 (b) 100 (c) 17 (d) 33.33 kmh^{-1}
5. P4L2V5Q10 (Multiple correct) A particle is projected in the XY-plane with $u = (2\hat{i} + 3\hat{j}) \text{ ms}^{-1}$ at $t = 0$ from the origin; acceleration $a = \hat{i} + 4\hat{j}$. (a) path is parabolic (b) path is a straight line (c) position at $t=1\text{s}$ is $(2.5\hat{i} + 5\hat{j}) \text{ m}$ (d) speed at $t=1\text{s}$ is $\sqrt{58} \text{ ms}^{-1}$
6. P4L9V3Q2 A plane flying horizontally at 98 m/s releases an object which reaches the ground in 10 s. Find the angle it makes with the horizontal on hitting the ground. (a) 55° (b) 45° (c) 60° (d) 75°
7. P4L11V5Q1 Boy throws ball upwards with $v_0 = 20 \text{ m/s}$. Wind imparts horizontal acceleration 4 m/s^2 to the left. Angle θ so ball returns to boy's hand ($g = 10 \text{ m/s}^2$). (a) $\tan^{-1}(1.2)$ (b) $\tan^{-1}(0.2)$ (c) $\tan^{-1}(2)$ (d) $\tan^{-1}(0.4)$

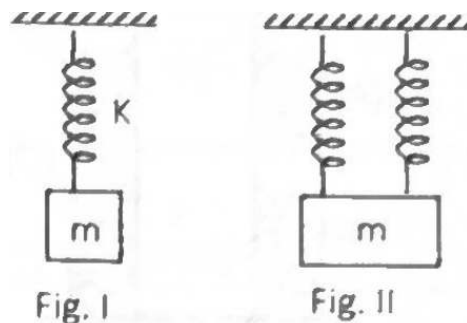


8. P6L1V3Q4 A block has a 15 N applied force and 5 N friction; mass 2 kg. Find net force and acceleration. (a) 5 N, 2.5 m/s^2 (b) 10 N, 5 m/s^2 (c) 20 N, 10 m/s^2 (d) 25 N, 12.5 m/s^2

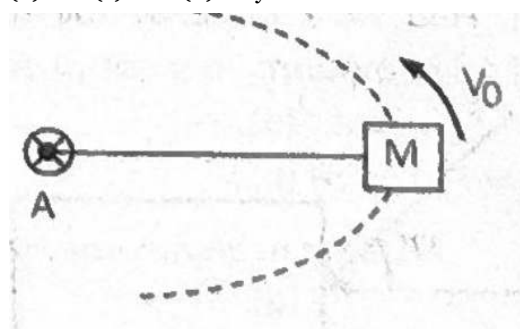
9. P6L5V3Q2 Extension in the spring equals ($K = 5000 \text{ N/m}$, $m = 5 \text{ kg}$ on a movable pulley): 0.04 / 0.02 / 0.01 / 0.08 m



10. P6L6V4Q1 A block is hung using a spring; spring cut into 2 equal parts and block re-hung with both in parallel (Fig. II). Ratio of extension Fig. I : one spring in Fig. II. (a) 4:1 (b) 1:4 (c) 1:1 (d) 2:1

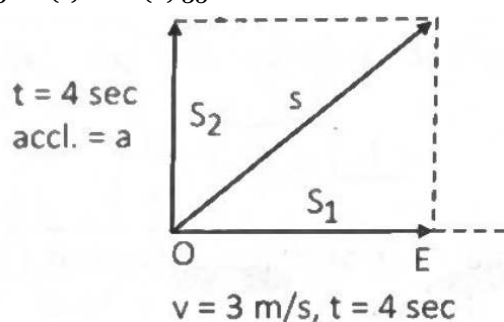


11. P7L1V3Q2 A 10 kg block is on a 30° inclined plane. If $\mu = 0.2$, what is the net force acting on it down the plane? (Take $g = 10 \text{ m/s}^2$): (a) 33 N (b) 40 N (c) 30 N (d) 25 N.
12. P7L1V6Q4 A block B of mass 2 kg rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.7. The frictional force on the block is: (a) 9.8 N (b) $0.7 \times 9.8\sqrt{3} \text{ N}$ (c) $9.8 \times 7 \text{ N}$ (d) $0.8 \times 9.8 \text{ N}$.
13. P8L1V3Q5 A block of mass m is pivoted with the help of a string at end A. This combination is free to rotate around A (the block swings in a circle of radius = string length, with speed v_0). Find the nature of the work done by the tension. (a) +ve (b) -ve (c) zero (d) may be +ve or -ve

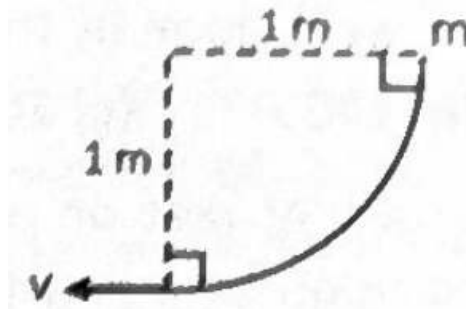


14. P8L2V4Q7 A block is constrained to move along x-axis under a force $F = -(2x) \text{ N}$. Find the workdone by this force when the block is displaced from $x = 2 \text{ m}$ to $x = -4 \text{ m}$? (a) 12 J (b) 8 J (c) -12 J (d) -8 J
15. P8L3V5Q5 A small block of mass m is kept on a rough inclined surface of inclination θ fixed in an elevator. The elevator goes up with a uniform velocity v and the block does not slide on the wedge. The work done by the force of friction on the block in time t will be (a) zero (b) $mgvt \cos^2\theta$ (c) $mgvt \sin^2\theta$ (d) $mgvt \sin 2\theta$

16. P8L8V3Q5 A crane is powered by a motor delivering 20 kW. The crane is raising a load of 5 tonnes. If $g = 10 \text{ ms}^{-2}$, the maximum speed with which the load can be lifted is (a) 40 ms^{-1} (b) 4 ms^{-1} (c) 0.04 ms^{-1} (d) 0.4 ms^{-1}
17. P4L3V3Q2 A stone dropped from the top of a tower is found to travel $\frac{5}{9}$ of the height of the tower during the last second of its fall. The time of fall is: (a) 2 s (b) 3 s (c) 4 s (d) 5 s
18. P6L2V4Q5 Body 2 kg, initial velocity 3 m/s along OE, force 4 N perpendicular to OE; distance from O after 4 s. (a) 10m (b) 30m (c) 20m (d) 35m

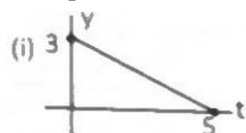


19. P7L1V2Q1 A block A of mass 4 kg is kept on the ground. The coefficient of friction between the block and the ground is 0.8. An external force of magnitude 30 N is applied parallel to the ground. The resultant force exerted by the ground on the block is ($g = 10 \text{ m/s}^2$): (a) 40 N (b) 30 N (c) zero (d) 50 N.
20. P8L4V3Q2 A block of mass 1 kg slides down a curved track which forms one quadrant of a circle of radius 1 m as shown in the figure. The speed of the block at the bottom of the track is $v = 2 \text{ m s}^{-1}$. The work done by the force of friction is (write the value)



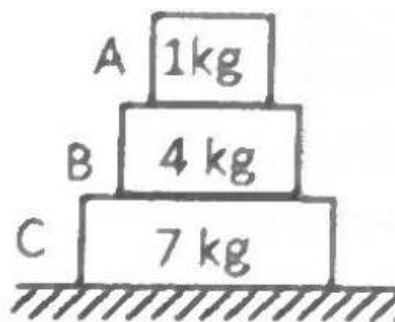
PART B · Q21–25 · NUMERICAL VALUE

21. P1L2V4Q3 Find slope of line in the following figure: (i) line from y-intercept 3 falling to intercept 5 on the horizontal axis (labelled t); (ii) line through x-intercept -3 rising to y-intercept 5



22. P4L11V4Q5 Particle A moves North with acceleration 5 ms^{-2} ; particle B moves North-East with acceleration $5\sqrt{2} \text{ ms}^{-2}$. Find relative acceleration of A w.r.t. B towards west.

23. P5L1VSuggQ3 Three blocks stacked A=1 kg, B=4 kg, C=7 kg (A on top, C on ground); find normal reaction between C & the ground, in newton



24. P8L4V3Q1 A block of mass 30.0 kg is being brought down by a chain. If the block acquires a speed of 40.0 cm/s in dropping down 2.00 m, find the work done (in J) by the chain during the process. (Take $g = 10 \text{ m/s}^2$)
25. P3L3L4V3Q4 Force inclined at 53° to horizontal, horizontal component 30 N: find the force.

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C2L2V3Q3 Light $\lambda = 500 \text{ nm}$ — energy of one photon. (a) $3.98 \times 10^{-19} \text{ J}$ (b) $2.48 \times 10^{-19} \text{ J}$ (c) $6.00 \times 10^{-19} \text{ J}$ (d) $1.33 \times 10^{-19} \text{ J}$
2. C2L4V4Q5 Coulomb force in 1st orbit is F — force in the 3rd orbit? (a) $F/9$ (b) $81F$ (c) $9F$ (d) $F/81$
3. C2L7V5Q3 Electron falls $n=5 \rightarrow 2$ — region observed & which series. (a) Visible, Balmer (b) Ultraviolet, Lyman (c) Infrared, Paschen (d) Infrared, Bracket
4. C3L1V3Q2 Average atomic mass of chlorine: 75.77% of 35 u, 24.23% of 37 u. (a) 35.45 u (b) 36.00 u (c) 35.00 u (d) 34.50 u
5. C3L4V6Q10 (Multiple correct) Correct statements about empirical / molecular / structural formula?
6. C3L11V3Q1 Multiply 2.5 (2 sig. fig) $\times 3.42$ (3 sig. fig). Round to correct sig. figs. (a) 8.6 (b) 8.55 (c) 8.550 (d) 8.4
7. C3L11V4Q4 Given the numbers 161 cm; 0.161 cm, 0.0161 cm. The number of significant figures for the three numbers is (a) 3, 4, 5 (b) 3, 3, 3 (c) 3, 3, 4 (d) 3, 4, 4

8. C3L12V5Q10 12 g carbon combine with 64 g sulphur to form CS_2 . 12 g carbon also combine with 32 g oxygen to form CO_2 . 10 g sulphur combine with 10 g oxygen to form SO_2 . These data illustrate the (a) law of multiple proportions (b) law of definite proportions (c) law of reciprocal proportions (d) law of gaseous volumes
9. C4L1V3Q1 An element has atomic number 17. What is its group and period? (a) Group 17, Period 3 (b) Group 7, Period 2 (c) Group 16, Period 4 (d) Group 3, Period 1
10. C4L1V6Q3 Elements A, B, C, D and E have the configurations A: $1s^2 2s^2 2p^1$; B: $1s^2 2s^2 2p^6 3s^2 3p^1$; C: $\dots 3s^2 3p^3$; D: $\dots 3s^2 3p^5$; E: $\dots 3s^2 3p^6 4s^2$. Which among these belong to the same group? A: $1s^2 2s^2 2p^1$; B: $1s^2 2s^2 2p^6 3s^2 3p^1$; C: $1s^2 2s^2 2p^6 3s^2 3p^3$; D: $1s^2 2s^2 2p^6 3s^2 3p^5$; E: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$. (a) A and B (b) C and D (c) B and C (d) D and E
11. C5L2V5Q1 According to Fajan's rule, polarization is more when: (a) small cation and large anion (b) small cation and small anion (c) large cation and large anion (d) large cation and small anion

12. C5L4V2Q2 For $Q = 3$ (Q = total number of orbitals), determine hybridisation. (a) sp (b) sp^2 (c) sp^3 (d) sp^3d
13. C5L8V3Q2 Find bond order and magnetism for NO (15 electrons). (a) 2.5, Paramagnetic (b) 2, Diamagnetic (c) 3, Diamagnetic (d) 1.5, Paramagnetic
14. C5L8V4Q3 According to molecular orbital theory, which of the following will not exist? (a) H_2^+ (b) Be_2 (c) B_2 (d) C_2
15. C5L9V5Q10 Arrange the following in order of increasing dipole moment: H_2O , H_2S , BF_3 . (a) $BF_3 < H_2S < H_2O$ (b) $H_2S < BF_3 < H_2O$ (c) $H_2O < H_2S < BF_3$ (d) $BF_3 < H_2O < H_2S$
16. C2L1V3Q3 Neutrons twice electrons, mass 54, neutral — number of protons? (a) 18 (b) 20 (c) 21 (d) 22
17. C3L3V4Q5 A compound has a haemoglobin-like structure. It has one atom of Fe and contains 4.6% Fe. The approximate molecular mass is (a) 100 (b) 1200 (c) 1400 (d) 1600 $g\ mol^{-1}$
18. C4L2V3Q3 An atom has $Z = 16$ and screening constant $\sigma = 12.5$. What is Z^* ? (a) 3.5 (b) 4.5 (c) 2.5 (d) 5.5
19. C5L2V2Q1 Calculate the formal charge on nitrogen in NH_4^+ (valence electrons = 5, lone pairs = 0, bonding electrons = 8). (a) 0 (b) -1 (c) +1 (d) +2
20. C2L1V3Q2 Isotope Y: mass 37, 20 neutrons — which statement is true? (a) Atomic number of Y is 17 and it is a halogen. (b) Y is a noble gas with 17 electrons. (c) Y must have 20 electrons and 17 protons. (d) Mass number and neutron number suggest it's an alkali metal.

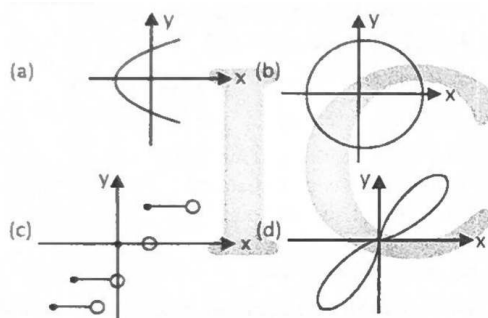
PART B · Q21–25 · NUMERICAL VALUE

21. C1L6V3Q5 Average kinetic energy of 5 moles of gas at $0^\circ C$. (a) $1.13 \times 10^4\ J$ (b) $1.42 \times 10^4\ J$ (c) $1.70 \times 10^4\ J$ (d) $2.26 \times 10^4\ J$
22. C1L1V3Q4 Convert 2 atm to mm Hg and pascal. (a) 1520 mmHg, 202650 Pa (b) 760 mmHg, 101325 Pa (c) 380 mmHg, 50662.5 Pa (d) 2000 mmHg, 101325 Pa
23. C1L2V4Q6 1.0 L flask of air $27^\circ C \rightarrow 37^\circ C$, P const — volume of air that escapes. (a) 28 mL (b) 33 mL (c) 37 mL (d) 40 mL
24. C1L4V4Q10 Dalton's law is NOT valid for which mixture? ($NH_3 + HCl$). (a) Oxygen and nitrogen (b) Ammonia and hydrogen chloride (c) Helium and hydrogen (d) Argon and oxygen
25. C1L4V5Q9 12 L flask $35^\circ C$: 0.3 mol CO_2 , 0.4 mol He, 0.5 mol N_2 — partial & total pressure. (a) $P(CO_2)=0.6\ atm$, $P(He)=0.8\ atm$, $P(N_2)=1.0\ atm$, $P_{total}=2.4\ atm$ (b) $P(CO_2)=0.9\ atm$, $P(He)=1.2\ atm$, $P(N_2)=1.5\ atm$, $P_{total}=3.6\ atm$ (c) $P(CO_2)=0.5\ atm$, $P(He)=0.7\ atm$, $P(N_2)=0.9\ atm$, $P_{total}=2.1\ atm$ (d) None of these

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L2V4Q5 Circular wire $r = 7\ cm$ bent into arc of $r = 12\ cm$; angle at centre. (a) 50° (b) 210° (c) 100° (d) 60°
2. M2L1V2Q4 $A = \{x : x^2 + 4 = 0, x \in R\}$; cardinality of $P(A)$. (a) 2 (b) 3 (c) 4 (d) None
3. M3L2V3Q3 $4a^2 \sin^2(3\pi/4) - 3[a \cdot \tan 225^\circ]^2 + [2a \cdot \cos 315^\circ]^2$. (a) 0 (b) a (c) $\sqrt{2}a$ (d) a^2
4. M4L1V3Q4 Which graph is the graph of a function?



5. M4L4V2Q4 Maximum value of $y = -\cos x$. (a) 0 (b) -1 (c) 1 (d) $-\pi$
6. M5AL2V4Q8 $(\cos 9^\circ + \sin 9^\circ)/(\cos 9^\circ - \sin 9^\circ) = \tan x$, then $x = ?$ (a) 45° (b) 54° (c) 27° (d) 18°
7. M5AL2V6Q7 $0 < A, B < \pi/4$, $\cos(A+B) = 4/5$, $\sin(A-B) = 5/13$, then $\tan 2A = ?$ (a) $16/63$ (b) $56/33$ (c) $1/65$ (d) $45/77$
8. M5BL2V4Q6 If $\tan(\theta_1) \cot(\theta_2) = k$, then the value of $\sin(\theta_1 + \theta_2)/\sin(\theta_1 - \theta_2) =$ (a) $(k-1)/(k+1)$ (b) $(k+1)/(k-1)$ (c) $(1-k)/(1+k)$ (d) $(1+k)/(1-k)$

PART B · Q21–25 · NUMERICAL VALUE

21. M5AL1V3Q1 $(\tan 75^\circ - \tan 60^\circ)/(1 + \tan 75^\circ \tan 60^\circ) = x$, then $(x + \sqrt{3})^2 = ?$
22. M6L2V4Q4 If $11 - 5x > -4$ and $4x + 13 \geq -11$, then $x \in [a, b]$; find $|a| + b$.

9. M5CL3V3Q2 $(\sin 3\theta - \cos 3\theta)/(\sin \theta + \cos \theta) + 1 =$ (a) $2\sin 2\theta$ (b) $2\cos 2\theta$ (c) $\tan 2\theta$ (d) $\cot 2\theta$
10. M6L1V3Q1 If $3(x-1) \leq 2(x-3)$ then 'x' is: (a) $(-\infty, -3]$ (b) $[3, \infty)$ (c) $[5, \infty)$ (d) $[0, \infty)$
11. M6L2V2Q3 Check if (2, 1) satisfies $x + 2y \geq 5$. (a) Yes (b) No (c) Sometimes (d) Only if $x = y$
12. M7L2V3Q3 If $a = (1/2)^{1/3}$ and $b = (1/2)^{1/2}$, then (a) $a = b$ (b) $a > b$ (c) $a < b$ (d) none
13. M7L3V4Q5 If $a^2 + b^2 - c^2 = 0$, then $1/\log_{-}(c+a) b + 1/\log_{-}(c-a) b =$ (a) 1 (b) 2 (c) -1 (d) -2
14. M8BL2V4Q3 If s is the sum of an infinite G.P., the first term a then the common ratio r given by (a) $(a-s)/s$ (b) $(s-a)/s$ (c) $a/(1-s)$ (d) $(s-a)/a$
15. M8CL1V5SuggQ4 Find the 4th term of an H.P. whose 7th term is $1/20$ and 13th term is $1/38$. (write the value)
16. M4L1V3Q2 Piecewise $f : x^2 + 1 (x \leq 0), 2x - 1 (0 < x < 5) \rightarrow f(-3) - f(2)$. (a) 7 (b) 10 (c) 4 (d) 5
17. M8AL1V3Q5 Find the 20th term from the last in the A.P. 3, 7, 11, ..., 399. (write the value)
18. M8BL3V5Q1 $1/2 + 1/4 + 1/8 + 1/16 + \dots \infty$ terms = x , $1/3 + 1/9 + 1/27 + \dots \infty$ terms = y , then the value of $x + 2y$. (write the value)
19. M8AL1V3Q3 Which term of the A.P. 3, 8, 13, ... is 248? (write the value)
20. M8BL1V3Q1 Which term of the sequence $\sqrt{2}, 2, 2\sqrt{2}, \dots$ is 1024 (write the value)

23. M2L1V3Q4 $A = \{x : x^2 = 16 \text{ and } 2x = 16\}$; elements n ; find 5^n
24. M3L1V4Q3 $(\sec A \sec B + \tan A \tan B)^2 - (\sec A \tan B + \tan A \sec B)^2$
25. M8AL1V3Q2 Find the 15th term of an A.P. 1, 4, 7, ...